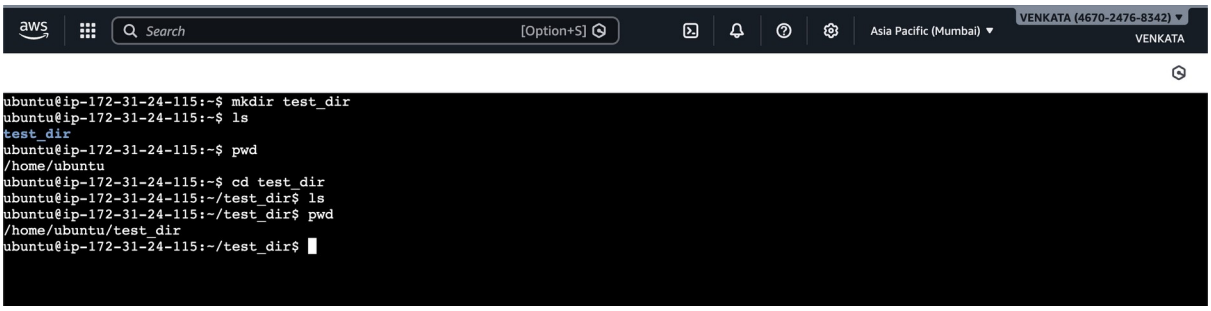
1. Creating and Renaming Files/Directories

Create a directory named test_dir using mkdir.



CMD : mkdir test_dir

Inside test_dir, create an empty file called example.txt.

```
ubuntu@ip-172-31-24-115:~$ mkdir test_dir
ubuntu@ip-172-31-24-115:~$ ls
test_dir
ubuntu@ip-172-31-24-115:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-24-115:~$ cd test_dir
ubuntu@ip-172-31-24-115:~/test_dir$ ls
ubuntu@ip-172-31-24-115:~/test_dir$ pwd
/home/ubuntu/test_dir
ubuntu@ip-172-31-24-115:~/test_dir$ touch example.txt
ubuntu@ip-172-31-24-115:~/test_dir$ ls
example.txt
ubuntu@ip-172-31-24-115:~/test_dir$ pwd
/home/ubuntu/test_dir
```

CMD : touch example.txt Rename example.txt to renamed_example.txt using mv

```
ubuntu@ip-172-31-24-115:~$ ls
test_dir
ubuntu@ip-172-31-24-115:~$ cd test_dir
ubuntu@ip-172-31-24-115:~/test_dir$ ls
example.txt
ubuntu@ip-172-31-24-115:~/test_dir$ mv example.txt rename_example.txt
ubuntu@ip-172-31-24-115:~/test_dir$ lst
Command 'lst' not found, but there are 23 similar ones.
ubuntu@ip-172-31-24-115:~/test_dir$ ls
rename_example.txt
ubuntu@ip-172-31-24-115:~/test_dir$
```

CMD: mv example.txt rename_example.txt

2. Viewing File Contents

Use cat to display the contents of /etc/passwd.

```
ubuntu@ip-172-31-24-115:~/etc/passwd$ cat sample.py

import random
import sys

# ---------------------------
# Dungeon Adventure Game
# ---------------------------

class Player:
    def __init__(self, name):
        self.name = name
        self.hp = 100
        self.gold = 0
        self.inventory = []

    def is_alive(self):
        return self.hp > 0

    def attack(self, enemy):
        damage = random.randint(5, 15)
        enemy.hp -= damage
        print(f"{self.name} attacks {enemy.name} for {damage} damage!")

    def heal(self):
        if "Potion" in self.inventory:
            self.hp += 20
            self.inventory.remove("Potion")
            print(f"{self.name} drinks a potion and heals 20 HP!")
        else:
            print("No potion available!")

class Enemy:
    def __init__(self, name, hp, attack_power):
```

CMD : cat sample.py

Display only the first 5 lines of /etc/passwd using head.

```
ubuntu@ip-172-31-24-115:~/etc/passwd$ head 5 sample.py
head: cannot open '5' for reading: No such file or directory
==> sample.py <==

import random
import sys

# --------------------------
# Dungeon Adventure Game
# --------------------------

class Player:
    def __init__(self, name):
ubuntu@ip-172-31-24-115:~/etc/passwd$
```

CMD : head 5 sample.py

Display only the last 5 lines of /etc/passwd using tail.

```
ubuntu@ip-172-31-24-115:~/etc/passwd$ tail 5 sample.py
tail: cannot open '5' for reading: No such file or directory
==> sample.py <==
            break
        room.enter(player)

    if player.is_alive():
        print("\nCongratulations! You survived the dungeon!")
        print(f"Final HP: {player.hp}, Gold: {player.gold}, Inventory: {player.inventory}")

if __name__ == "__main__":
    main()

ubuntu@ip-172-31-24-115:~/etc/passwd$
```

CMD : tail 5 sample.py

3.Searching for Patterns

Use grep to find all lines containing the word "root" in /etc/passwd.

```
ubuntu@ip-172-31-24-115:~/7Q$ cd ..
ubuntu@ip-172-31-24-115:~$ mkdir 3Q
ubuntu@ip-172-31-24-115:~$ ls
3Q  6Q  7Q  etc  index.html  test_dir
ubuntu@ip-172-31-24-115:~$ cd 3Q
ubuntu@ip-172-31-24-115:~/3Q$ grep root /etc/passwd
root:x:0:0:root:/root:/bin/bash
ubuntu@ip-172-31-24-115:~/3Q$
```

CMD : grep root /etc/passwd

EXPLANATION :

`root` → username

`x` → password placeholder (actual password stored in `/etc/shadow`)

`0:0` → user ID (UID) and group ID (GID), both 0 for root

`root` → user description

`/root` → home directory

`/bin/bash` → login shell

4. Zipping and Unzipping

Compress the test_dir directory into a file named test_dir.zip using zip.

Unzip test_dir.zip into a new directory named unzipped_dir.

```
ubuntu@ip-172-31-24-115:~/4a$ ls -al
total 12
drwxrwxr-x  3 ubuntu ubuntu 4096 Feb 16 05:48 .
drwxr-x--- 10 ubuntu ubuntu 4096 Feb 16 05:46 ..
drwxrwxr-x  2 ubuntu ubuntu 4096 Feb 16 05:48 test_dir
ubuntu@ip-172-31-24-115:~/4a$ zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
ubuntu@ip-172-31-24-115:~/4a$ ls -al
total 16
drwxrwxr-x  3 ubuntu ubuntu 4096 Feb 16 05:48 .
drwxr-x--- 10 ubuntu ubuntu 4096 Feb 16 05:46 ..
drwxrwxr-x  2 ubuntu ubuntu 4096 Feb 16 05:48 test_dir
-rw-rw-r--  1 ubuntu ubuntu  168 Feb 16 05:48 test_dir.zip
ubuntu@ip-172-31-24-115:~/4a$ unzip test_dir.zip -d unzipped_dir
Archive:  test_dir.zip
   creating: unzipped_dir/test_dir/
ubuntu@ip-172-31-24-115:~/4a$ ls -al
total 20
drwxrwxr-x  4 ubuntu ubuntu 4096 Feb 16 05:49 .
drwxr-x--- 10 ubuntu ubuntu 4096 Feb 16 05:46 ..
drwxrwxr-x  2 ubuntu ubuntu 4096 Feb 16 05:48 test_dir
-rw-rw-r--  1 ubuntu ubuntu  168 Feb 16 05:48 test_dir.zip
drwxrwxr-x  3 ubuntu ubuntu 4096 Feb 16 05:49 unzipped_dir
ubuntu@ip-172-31-24-115:~/4a$
```

FOR ZIPCMD : zip -r test_dir.zip test_dirEXPLANATION :

`zip` → the compression utility

`-r` → recursive, so it includes all subdirectories and files

`test_dir.zip` → the name of the output archive

`test_dir` → the directory you want to compress

FOR UNZIPCMD : unzip test_dir.zip -d unzipped_dir

EXPLANATION:

`unzip` → extracts the archive

- `test_dir.zip` → the archive you created
- `-d unzipped_dir` → specifies the destination directory

If `unzipped_dir` doesn’t exist, `unzip` will create it automatically

5. Downloading Files

Use wget to download a file from a URL (e.g., https://example.com/sample.txt).

```
ubuntu@ip-172-31-24-115:~$ wget https://example.com/
--2026-02-16 05:11:39--  https://example.com/
Resolving example.com (example.com)... 104.18.27.120, 104.18.26.120, 2606:4700::6812:1a78, ...
Connecting to example.com (example.com)|104.18.27.120|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: ‘index.html’

index.html                              [ <=>                                                                   ]     528  --.-KB/s    in 0s

2026-02-16 05:11:39 (207 MB/s) - ‘index.html’ saved [528]

ubuntu@ip-172-31-24-115:~$ ls -al
total 48
drwxr-x--- 6 ubuntu ubuntu 4096 Feb 16 05:11 .
drwxr-xr-x 3 root   root   4096 Feb 14 11:57 ..
-rw------- 1 ubuntu ubuntu 5536 Feb 16 05:07 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Mar 31  2024 .bashrc
drwx------ 2 ubuntu ubuntu 4096 Feb 14 11:58 .cache
-rw-r--r-- 1 ubuntu ubuntu  807 Mar 31  2024 .profile
drwx------ 2 ubuntu ubuntu 4096 Feb 14 11:57 .ssh
-rw-r--r-- 1 ubuntu ubuntu    0 Feb 14 23:03 .sudo_as_admin_successful
drwxrwxr-x 3 ubuntu ubuntu 4096 Feb 14 15:21 etc
-rw-rw-r-- 1 ubuntu ubuntu  528 Feb 12 14:00 index.html
drwxrwxr-x 2 ubuntu ubuntu 4096 Feb 16 04:59 test_dir
ubuntu@ip-172-31-24-115:~$
```

CMD : wget https://example.com/

6. Changing Permissions

Create a file named secure.txt and change its permissions to read-only for everyone using chmod.

```
ubuntu@ip-172-31-24-115:~/6Q$ touch secure.txt
ubuntu@ip-172-31-24-115:~/6Q$ ls -la
total 8
drwxrwxr-x 2 ubuntu ubuntu 4096 Feb 16 05:14 .
drwxr-x--- 7 ubuntu ubuntu 4096 Feb 16 05:13 ..
-rw-rw-r-- 1 ubuntu ubuntu    0 Feb 16 05:14 secure.txt
ubuntu@ip-172-31-24-115:~/6Q$ chmod 444 secure.txt
ubuntu@ip-172-31-24-115:~/6Q$ ls -la
total 8
drwxrwxr-x 2 ubuntu ubuntu 4096 Feb 16 05:14 .
drwxr-x--- 7 ubuntu ubuntu 4096 Feb 16 05:13 ..
-r--r--r-- 1 ubuntu ubuntu    0 Feb 16 05:14 secure.txt
ubuntu@ip-172-31-24-115:~/6Q$
```

CMD : chmod 444 secure.txt

7. Working with Environment Variables

Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!".

```
ubuntu@ip-172-31-24-115:~/7Q$ export MY_VAR="Hello, Linux!"
ubuntu@ip-172-31-24-115:~/7Q$ ls -la
total 8
drwxrwxr-x 2 ubuntu ubuntu 4096 Feb 16 05:19 .
drwxr-x--- 8 ubuntu ubuntu 4096 Feb 16 05:19 ..
ubuntu@ip-172-31-24-115:~/7Q$ echo MY_VAR
MY_VAR
ubuntu@ip-172-31-24-115:~/7Q$ echo $MY_VAR
Hello, Linux!
ubuntu@ip-172-31-24-115:~/7Q$
```

CMD : export MY_VAR=“Hello, Linux!”

CMD : echo $MY_VAR

Submission Guidelines -: Attach Screenshots or command along with explanation and submit in doc(google doc or microsoft doc) format also attach github repo link